

Literatures

- Problem Statement:

1. Signal Integrity Design for High Speed Digital Circuits: Progress and Directions - 1 **.Signal Integrity Design for High Speed Digital Circuits Progress and Directions**

URL: [Signal Integrity Design for High-Speed Digital Circuits: Progress and Directions | IEEE Journals & Magazine | IEEE Xplore](#)

Authors: J. Fan, X. Ye, J. Kim, B. Archambeault and A. Orlandi, "Signal Integrity Design for High Speed Digital Circuits: Progress and Directions," in IEEE Transactions on Electromagnetic Compatibility, vol. 52, no. 2, pp. 392-400, May 2010, doi: 10.1109/TEM.2010.2045381.

Keywords: [Transmission line measurements;Silicon;Integrated circuit modeling;Time domain analysis;Noise;Jitter;Digital circuits;Differential signaling;discontinuities;equalization;equivalent circuit modeling;high-speed digital circuits;jitter analysis;mode conversion;modulation;signal integrity (SI);SI measurements;transmission line losses]

2. Overview of Signal Integrity and EMC Design Technologies on PCB: Fundamentals and Latest Progress. - 2. Overview of Signal Integrity and EMC Design Technologies on PCB Fundamentals and Latest Progress

URL: [Overview of Signal Integrity and EMC Design Technologies on PCB: Fundamentals and Latest Progress | IEEE Journals & Magazine | IEEE Xplore](#)

Authors: T. -L. Wu, F. Buesink and F. Canavero, "Overview of Signal Integrity and EMC Design Technologies on PCB: Fundamentals and Latest Progress," in IEEE Transactions on Electromagnetic Compatibility, vol. 55, no. 4, pp. 624-638, Aug. 2013, doi: 10.1109/TEM.2013.2257796.

Keywords: [Electromagnetic compatibility;Noise;Crosstalk;Integrated circuit modeling;Silicon;Slot lines;Scattering parameters;Circuit demonstrator;crosstalk;differential signaling;eye diagram;macromodeling;mode conversion;paramter variability;PCB;polynomial chaos;practical training;signal integrity;TDR;trace discontinuity]

3. Impact of pcb manufacturing process variations on trace impedance - 3. Impact of pcb manufacturing process variations on trace impedance

URL:

https://www.researchgate.net/publication/235330265_Impact_of_PCB_manufacturing_process_variations_on_trace_impedance

Authors: Renbi, A.; Carlson, J.; Delsing, J. Impact of pcb manufacturing process variations on trace impedance. In Proceedings of the International Symposium on Microelectronics, Long Beach, CA, USA, 9–13 October 2011; International Microelectronics As-sembly and Packaging Society: Pittsburgh, PA, USA, 2008; pp. 000891–000895

- ML in SI applications:

5. Efficient Synthesis and Simulation of High-Density Interconnects Using Machine Learning - 5. Efficient Synthesis and Simulation of High-Density Interconnects Using Machine Learning

URL: [Efficient Synthesis and Simulation of High-Density Interconnects Using Machine Learning | IEEE Conference Publication | IEEE Xplore](#)

Authors: D. Sreekumar and S. Gupta, "Efficient Synthesis and Simulation of High-Density Interconnects Using Machine Learning," 2025 IEEE 29th Workshop on Signal and Power Integrity (SPI), Gaeta, Italy, 2025, pp. 1-4, doi: 10.1109/SPI64682.2025.11014451.

Keywords: [Training;Accuracy;Computational modeling;Stripline;Machine learning;Predictive models;Data models;Scattering parameters;Electromagnetics;Signal integrity;ANN;Ensemble modelling;high-speed highdensity interconnects;machine-learning;random distribution;signal integrity;S-parameters]

- Physics-Informed Modelling

7. Modeling Electrically Long Interconnects Using Physics-Informed Delayed Gaussian Processes - 7. Modeling Electrically Long Interconnects Using Physics Informed Delayed Gaussian Processes

URL: [Modeling Electrically Long Interconnects Using Physics-Informed Delayed Gaussian Processes | IEEE Journals & Magazine | IEEE Xplore](#)

Authors: F. Garbuglia, T. Reuschel, C. Schuster, D. Deschrijver, T. Dhaene and D. Spina, "Modeling Electrically Long Interconnects Using Physics-Informed Delayed Gaussian Processes," in IEEE Transactions on Electromagnetic Compatibility, vol. 65, no. 6, pp. 1715-1723, Dec. 2023, doi: 10.1109/TEMPC.2023.3317917.

Keywords: [Kernel;Scattering parameters;Data models;Transforms;Propagation delay;Gaussian processes;Machine learning;Delay estimation;Gabor transform;Gaussian processes (GP);interconnects;kernels;machine learning (ML);S-parameters]

- State-of-the-art for Forward models

6.Surrogate Modeling With Complex-Valued Neural Nets for Signal Integrity Applications - 6.Surrogate Modeling With Complex-Valued Neural Nets for Signal Integrity Applications

URL: [Surrogate Modeling With Complex-Valued Neural Nets for Signal Integrity Applications | IEEE Journals & Magazine | IEEE Xplore](#)

Authors: O. Akinwande, S. Erdogan, R. Kumar and M. Swaminathan, "Surrogate Modeling With Complex-Valued Neural Nets for Signal Integrity Applications," in IEEE Transactions on Microwave Theory and Techniques, vol. 72, no. 1, pp. 478-489, Jan. 2024, doi: 10.1109/TMTT.2023.3319835.

Keywords: [Scattering parameters;Integrated circuit modeling;Artificial neural networks;Computational modeling;Predictive models;Behavioral sciences;Data models;Design space exploration;inverse design;neural networks (NNs);packaging;passivity;signal integrity (SI);surrogate modeling]

4. Signal Integrity Analysis and Design Optimization using Neural Networks - 4. Signal Integrity Analysis and Design Optimization using Neural Networks

URL: [Signal Integrity Analysis and Design Optimization using Neural Networks | IEEE Conference Publication | IEEE Xplore](#)

Authors: J. Konduru, O. Mikulchenko, L. Y. Foo and J. E. Schutt-Ainé, "Signal Integrity Analysis and Design Optimization using Neural Networks," *2024 IEEE 74th Electronic Components and Technology Conference (ECTC)*, Denver, CO, USA, 2024, pp. 924-928, doi: 10.1109/ECTC51529.2024.00150.

Keywords: [Training;Performance evaluation;Adaptation models;Accuracy;Neural networks;Machine learning;Predictive models;neural networks;machine learning;signal and power integrity;high-speed channels;channel modeling]

- Inverse Tandem Model (Inspirations)

8. Channel Inverse Design Using Tandem Neural Network - 8. Channel Inverse Design Using Tandem Neural Network

URL: <https://ieeexplore.ieee.org/document/9874935>

Authors: H. Ma et al., "Channel Inverse Design Using Tandem Neural Network," 2022 IEEE 26th Workshop on Signal and Power Integrity (SPI), Siegen, Germany, 2022, pp. 1-3, doi: 10.1109/SPI54345.2022.9874935.

keywords: [Training;Performance evaluation;Conferences;Neurons;Artificial neural networks;Attenuation;Numerical models;high-speed link;inverse design;neural network;impedance and attenuation]

- cVAE

9. Learning Structured Output Representation using Deep Conditional Generative Models - 9. Learning Structured Output Representation using Deep Conditional Generative Models

URL:

https://papers.nips.cc/paper_files/paper/2015/hash/8d55a249e6baa5c06772297520da2051-Abstract.html

Authors: Kihyuk Sohn, Xinchun Yan, Honglak Lee, "Learning Structured Output Representation using Deep Conditional Generative Models", 2015, NIPS'15: Proceedings of the 29th International Conference on Neural Information Processing Systems - Volume 2, Pages 3483 - 3491

10. On-Demand Inverse Design for Narrowband Nanophotonic Structures Based on Generative Model and Tandem Network - 10. On-Demand Inverse Design for Narrowband Nanophotonic Structures Based on Generative Model and Tandem Network

URL: <https://arxiv.org/abs/2507.14761>

Authors: Yuxiao Li, Taeyoon Kim, Allen Zhang, Zengbo Wang, Yongmin Liu

<https://doi.org/10.48550/arXiv.2507.14761>

cVAE (a) . Leveraging generative neural networks for accurate, diverse, and robust nanoparticle design - Leveraging generative neural network for accurate, diverse, and robust nanoparticle design

URL: <https://pubmed.ncbi.nlm.nih.gov/39659763/>

Authors: Rahman T, Tahmid A, Arman SE, Ahmed T, Rakhy ZT, Das H, Rahman M, Azad AK, Wahadoszamen M, Habib A. Leveraging generative neural networks for accurate, diverse, and robust nanoparticle design. Nanoscale Adv. 2024 Dec 9;7(2):634-642. doi: 10.1039/d4na00859f. PMID: 39659763; PMCID: PMC11627239.

- Rational Functions (Inspiration)

11. Parametric Modeling of Microwave Components Using Adjoint Neural Networks and Pole-Residue Transfer Functions With EM Sensitivity Analysis - 11. Parametric Modeling of Microwave Components Using Adjoint Neural Networks and Pole-Residue Transfer Functions With EM Sensitivity Analysis

URL: <https://ieeexplore.ieee.org/document/7847393>

Authors: F. Feng, V. -M. -R. Gongal-Reddy, C. Zhang, J. Ma and Q. -J. Zhang, "Parametric Modeling of Microwave Components Using Adjoint Neural Networks and Pole-Residue Transfer Functions With EM Sensitivity Analysis," in IEEE Transactions on Microwave Theory and Techniques, vol. 65, no. 6, pp. 1955-1975, June 2017, doi: 10.1109/TMTT.2017.2650904.

keywords: {Transfer functions;Neural networks;Parametric statistics;Training;Microwave theory and techniques;Data models;Sensitivity analysis;Electromagnetic (EM);neural network;parametric modeling;sensitivity analysis;transfer function}

12. Rational approximation of frequency domain responses by vector fitting - 12. Rational approximation of frequency domain responses by vector fitting

URL: [Rational approximation of frequency domain responses by vector fitting | IEEE Journals & Magazine | IEEE Xplore](#)

Authors: B. Gustavsen and A. Semlyen, "Rational approximation of frequency domain responses by vector fitting," in IEEE Transactions on Power Delivery, vol. 14, no. 3, pp. 1052-

1061, July 1999, doi: 10.1109/61.772353.

Keywords: [Frequency domain analysis;Function approximation;Power system modeling;Power system simulation;Resonance;Frequency dependence;Computational modeling;Power system dynamics;Convolutional codes;Frequency measurement]

Rational Layer (a). Rational neural networks Rational neural networks

URL: <https://arxiv.org/abs/2004.01902>

Authors: Nicolas Boullé, Yuji Nakatsukasa, Alex Townsend,
<https://doi.org/10.48550/arXiv.2004.01902>

- Jacobian Regularization

13. Robust Learning with Jacobian Regularization - 13. Robust Learning with Jacobian Regularization

URL: <https://arxiv.org/abs/1908.02729>

Authors: Judy Hoffman, Daniel A. Roberts, Sho Yaida,
<https://doi.org/10.48550/arXiv.1908.02729>(<https://doi.org/10.48550/arXiv.1908.02729>)

14. Deep Learning for Inverse Problems: Bounds and Regularizers - 14. Deep Learning for Inverse Problems Bounds and Regularizers

URL: (PDF) [Deep Learning for Inverse Problems: Bounds and Regularizers](#)

Authors: Amjad, Jaweria & Lyu, Zhaoyan & Rodrigues, M.R.D.. (2019). Deep Learning for Inverse Problems: Bounds and Regularizers. 10.48550/arXiv.1901.11352.

Jacobian (a). Jacobian Regularization Stabilizes Long-Term Integration of Neural Differential Equations

URL: <https://arxiv.org/abs/2602.04608>

Authors: Maya Janvier, Julien Salomon, Etienne Meunier,
<https://doi.org/10.48550/arXiv.2602.04608>(<https://doi.org/10.48550/arXiv.2602.04608>)

Jacobian (b). Computationally Efficient Design of an LNA Input Matching Network Using Automatic Differentiation

URL:

https://www.researchgate.net/publication/392389145_Computationally_Efficient_Design_of_an_LNA_Input_Matching_Network_Using_Automatic_Differentiation#fullTextFileContent

Authors: K. A. Shila, "Computationally Efficient Design of an LNA Input Matching Network Using Automatic Differentiation," in *IEEE Journal of Microwaves*, vol. 5, no. 4, pp. 972-982, July 2025, doi: 10.1109/JMW.2025.3568779.

keywords: [Noise;Optimization;Impedance matching;Finite difference methods;Machine learning;Stripline;Noise figure;Impedance;Computational efficiency;Transmission lines;Radio astronomy;Machine learning;non-uniform transmission lines;optimization;radio astronomy]

- Database for training

15. SI/PI-Database of PCB-Based Interconnects for Machine Learning Applications - 15. SIPI-Database of PCB-Based Interconnects for Machine Learning Applications

URL: <https://ieeexplore.ieee.org/document/9361755>

Authors: M. Schierholz *et al.*, "SI/PI-Database of PCB-Based Interconnects for Machine Learning Applications," in *IEEE Access*, vol. 9, pp. 34423-34432, 2021, doi: 10.1109/ACCESS.2021.3061788.

Keywords: [Tools;Databases;Electromagnetic compatibility;Integrated circuit interconnections;Machine learning;Electromagnetics;Simulation;Artificial neural network;database;electromagnetic compatibility;machine learning;power integrity;signal integrity]

16. openEMS – a free and open source equivalent-circuit (EC) FDTD simulation platform - 16. openEMS – a free and open source equivalent-circuit (EC) FDTD simulation platform

URL: [openEMS – a free and open source equivalent-circuit \(EC\) FDTD simulation platform supporting cylindrical coordinates suitable for the analysis of traveling wave MRI applications - Liebig - 2013 - International Journal of Numerical Modelling: Electronic Networks, Devices and Fields - Wiley Online Library](#)

Authors: Liebig, T., Rennings, A., Held, S. and Erni, D. (2013), openEMS – a free and open source equivalent-circuit (EC) FDTD simulation platform supporting cylindrical coordinates suitable for the analysis of traveling wave MRI applications. Int. J. Numer. Model., 26: 680-696. <https://doi.org/10.1002/jnm.1875>

K. A. Shila, "Computationally Efficient Design of an LNA Input Matching Network Using Automatic Differentiation," in IEEE Journal of Microwaves, vol. 5, no. 4, pp. 972-982, July 2025, doi: 10.1109/JMW.2025.3568779. keywords: {Noise;Optimization;Impedance matching;Finite difference methods;Machine learning;Stripline;Noise figure;Impedance;Computational efficiency;Transmission lines;Radio astronomy;Machine learning;non-uniform transmission lines;optimization;radio astronomy},
port-finding logic

Teledyne LeCroy, "Operator's Manual Eye Doctor II Software," which details the universal language of SI S-parameters and standard N -port block ordering for transmission line de-embedding.

Touchstone® File Format Specification, Version 2.1, Ratified by the IBIS Open Forum January 26, 2024

utils function references

Mixed-Mode S-Parameters: Bockelman, D. E., & Eisenstadt, W. R. (1995). Combined Differential and Common-Mode Scattering Parameters: Theory and Simulation. IEEE Transactions on Microwave Theory and Techniques, 43(7), 1530-1539. Link (This is the original paper that defined the 'M' matrix used in your code).

A. Ferrero and M. Pirola, "Generalized mixed-mode S-parameters," in IEEE Transactions on Microwave Theory and Techniques, vol. 54, no. 1, pp. 458-463, Jan. 2006, doi: 10.1109/TMTT.2005.860497.

Passivity and Causality: Triverio, P., Grivet-Talocia, S., Nakhla, M. S., Canavero, F. G., & Achar, R. (2007). Stability, Causality, and Passivity in Electrical Interconnect Models. IEEE Transactions on Advanced Packaging, 30(4), 795-808. Link (This paper justifies the $I - S^H S$ Eigenvalue check used to drop bad data).

Pozar, D. M. (2011). Microwave Engineering (4th ed.). John Wiley & Sons. (Note: Use this textbook as the standard citation for the $b = S a$ incident/reflected power wave derivations).
passivity

Reciprocity Theorem: Pozar, D. M. (2011). Microwave Engineering (4th ed.). John Wiley & Sons. (Chapter 4: Microwave Network Analysis). (This textbook justifies averaging the transpose matrix).

DC Extrapolation for Neural Networks: Achar, R., & Nakhla, M. S. (2001). Simulation of High-Speed Interconnects. Proceedings of the IEEE, 89(5), 693-728. [Link](#) (Explains why DC operating points are critical for macromodeling and time-domain transitions).